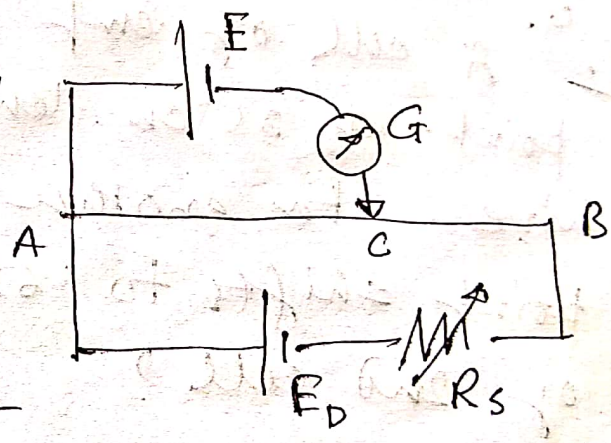
On Potentiometer

Q. 1 emf 'E' of cell shown in the fig has to be measured, by a potentiometer. The potentiometer wire AB has length '100 cm' and resistance '5Ω'. Emf of driver cell $E_D = 6V$ & series resistance $R_S = 5\Omega$. Null point is obtained at C with AC = 75 cm. What is E?



[2.25 V]

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Q.2 A 10 m long uniform metallic wire of resistance 20Ω is used as a potentiometer wire. This wire is connected in series with another resistance of 480 Ω and a battery of emf 5V having negligible internal resistance. If an unknown emf 'e' is balanced across '6m' of potentiometer wire, Calculate

(i) potential gradient across potentiometer wire. [0.02 V/m]

(ii) value of unknown emf e [0.12 volt]

Q.3 A cell gives a balance ~~length~~ with 85 cm of a potentiometer wire. When terminals of the cell is connected to a resistance of 7.5 Ω a balance is obtained at 75 cm. Find internal resistance of the cell. [1 Ω]

Q.4 A cell of emf 1.25 V gives a balance point at 35 cm length of a potentiometer wire. For another cell, the balance point shifts to 63 cm. What is emf of 2nd cell? [2.25 volt]

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5. A potentiometer wire is 100 cm long and a constant potential difference is maintained across it. Two batteries are connected in series first to support one another and then in opposite direction. The balance points are obtained at 50 cm and 10 cm from the +ve end of the wire in two cases. Find ratio of emfs? [3 : 2]

6. A potentiometer wire of length L and resistance r is connected in series with a battery of emf E_0 and a resistance r_1. An unknown emf 'E' is balanced at a length l of the potentiometer wire. Find E?

$$\left[\frac{E_0 r l}{(r+r_1)L} \right]$$

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